

Solution Requirements

Date	2 July 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely access the Smart Lender application through the web interface. If login is not implemented, access is limited to authorized users during deployment.	High
2	Authorization levels	Loan Applicants can submit loan details and view predictions. Credit Officers/Financial Analysts can evaluate applicant details and review prediction results.	High

3	External interfaces	The system interacts with the trained machine learning model (.pkl), dataset, Flask web application, and HTML pages.	High
4	Transactions processing	The system accepts applicant details, validates the input, processes the data through the ML model, and returns the loan eligibility prediction.	High
5	Reporting	The application displays the loan approval or rejection result. Prediction results can also be exported or stored for future analysis if required.	Medium
6	Business rules, etc.	Loan eligibility is determined based on applicant information such as income, employment status, education, credit history, loan amount, loan term, and property area using the trained XGBoost model.	High
7	Compliance to laws or regulations	Applicant information should be handled securely and confidentially, following applicable data privacy and security guidelines.	Medium

8	<i>Other</i>	The application should provide a simple, user-friendly interface for entering applicant information and viewing prediction results.	Medium
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Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should process applicant data and generate predictions quickly.	Prediction displayed within 2 seconds
2	Scalability	The system should support multiple users and increasing numbers of loan applications.	Supports 100+ users without performance degradation
3	Security & Data Privacy	Applicant data should be protected from unauthorized access and stored securely.	Secure handling of user data and input validation
4	Reliability & Availability	The application should operate consistently without unexpected failures.	99% uptime during deployment
5	Usability & Accessibility	The application should provide a simple, responsive, and easy-to-use interface.	User-friendly interface compatible with modern web browsers

6	<i>Other</i>	The system should be easy to update with new ML models or additional features in the future.	Modular Flask application with reusable code structure
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